

Predictive Analytics for Asthma Deterioration using Environmental and Clinical Factors

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Abstract

Asthma is the most common lung condition in the UK affecting around 8 million people, with 5.4 million receiving treatment for it (National Institute for Health and Care Excellence, 2023). Each year, asthma leads to about 60,000 hospital admissions and 200,000 bed days, and an average of 4 daily deaths (National Institute for Health and Care Excellence, 2025). Its financial costs are astronomical, with direct NHS costs estimated to be £1.3 billion, with indirect costs nearing £4.5 billion (National Institute for Health and Care Excellence, 2025). This coursework applies the CRISP-DM methodology to design, implement and evaluate a data mining workflow to predict worsening asthma symptoms in patients, based on demographic and environmental factors from a synthetic health dataset. Data familiarisation, preparation and feature engineering were performed in SQL to identify key predictors needed for model training. Three models were trained and evaluated in MATLAB, using accuracy, precision, sensitivity/recall and ROC/AUC metrics. Despite The Decision Tree and Logistic models achieving high accuracies (86.23%), both showed weaker discriminatory capabilities, with AUC values of 0.595 and 0.686 respectively. The Random Forest Model, however, recorded the lowest accuracy (61.8%) due to adjustments for class imbalance, however, it achieved the highest sensitivity (57.9%). Despite its lower AUC value (0.657), it was the most suitable model, as model sensitivity was prioritised over accuracy. In that context, incorrect prediction of asthma deterioration (false positives) can be managed by preventing an over-allocation of funds. However, false negatives pose a deadlier risk, since failure to detect worsening symptoms could delay treatment, putting patients in danger. The Random Forest model's sensitivity (57.9%) highlights its strong predictive ability to support timely clinical and cost based interventions. With proper threshold tuning and class weighting (He and Garcia, 2009), its impact could be enhanced. To maximise its performance, the model requires high volumes of data aggregated at the national level, not at localised GP practice levels, where data quality and exposure to more features are limited. Cloud based architectures present a deployment pathway, addressing data synchronisation, governance, privacy and the infrastructural challenges faced in managing large clinical datasets compiled from localised GPs.

1.0 Business Understanding

1.1 The Rationale for Predicting Asthma Deterioration

Predicting asthma deterioration enables early health recommendations for identified patients at risk of worsening symptoms. These predictions advise patients to modify their lifestyles to prevent the exacerbation of its symptoms, reducing hospitalisations and deaths. This consequently reduces the financial strain by the NHS in the management of Asthma.

The NHS proactively identifies at risk patients, by using routine primary-care data and identified environmental factors. This is done by integrating At-Risk Registers into primary care to stop Asthma crises (Health Research Authority, 2025). The registers enable clinicians to target high-risk individuals at each encounter, supporting early intervention to reduce severe exacerbations and improve outcomes (Smith et al., 2018). However, many asthma patients, despite NHS efforts, do not adhere to preventive therapy, consequently exacerbating its symptoms and leading to intervention, hospitalisations and deaths. Also, NHS patient data is siloed across GP practices, hospitals, and community care.

1.2 Benefits of National Level Analytics

The analysis investigates whether environmental conditions like motorway proximity, pollution and smoking cause higher asthma worsening rates due to the heightened exposure of traffic related pollutants. NHS healthcare data remains fragmented at the local level, across primary, secondary and community services thereby limiting access to unified patient histories. It constrains predictive modelling effectiveness and proactive risk identification (Kushi et al., 2024). Local variations in data quality and volume may yield unreliable predictive models resulting in unequal care across regions.

National level data mining at the national level enables aggregation and scalable processing of patient data across multiple healthcare locations. It supports complete, long term patient histories needed to manage asthma related risks. It also provides a vast exposure of necessary features to support high performance models to make meaningful predictions, insights difficult to obtain from isolated GP practice data.

2.0 Data Understanding (SQL Exploration)

2.1 The Dataset

The dataset is a synthetic patient dataset, generated using conventional SNOMED codes, to ensure clear definitions and consistency of its features. It contains 10000 rows and 37 columns where each record represents an individual patient. SQL inspection of the first 20 rows reveals the following:

```
select
*
from patients
fetch first 20 rows only;
```

Fig 1: SQL code for initial dataset inspection

- The ID column functions as a primary key which has no duplicates
- Binary data type for ASTHMA_WORSENERD, time series for date columns and object/integer types for GENDER.
- Variables encoded using SNOMED codes (eg. Gender, smok_code1 and other clinical measurements)

This requires a careful need for a thorough data preparation for model training.

2.2 Data Quality

The quality of the data was summarised across demographic compositions, clinical measurements, clinical codes and data ranges.

2.2.1 Demographic Composition

2.2.1.1 Gender Distribution

The SQL COUNT function revealed several inconsistent entries for Gender. SNOMED codes for Male (446141000124107) and Female (446151000124109) made up 40.27% and 39.73% of column data respectively. However, about 15% of entries used 'M', 'F' and 'O'.

```
1  select
2  gender,
3  COUNT(gender) as GenderCount
4  from patients
5  group by gender
6  order by gender asc
```

Fig 1: SQL code for GENDER column inspection

Gender Code	Count	Percentage
446141000124107 (Male)	4027	40.27%
446151000124109 (Female)	3973	39.73%
F	523	5.23%
O	501	5.01%
M	493	4.93%

Table 1: A summary of patient gender in the dataset

2.2.1.1 Year of Birth

The *YEAR_OF_BIRTH* variable shows even distribution across the birth years, with a concentration between the 1960s - 2000s. Using the SQL query below, the following descriptive statistics are observed:

```
select
AVG(year_of_birth) as Mean,
MEDIAN(year_of_birth) as Median,
MAX(year_of_birth) as Max,
MIN(year_of_birth) as Min
from patients
```

Fig 2: SQL code for Year of Birth inspection

- Range: 1920 - 2025.
- Mean: 1983 (average age 42).
- Median: 1986 (approximate age of 39)

Births recorded in 2025 suggest the need for age credibility checks at the data preparation phase to avoid a distortion of patterns models may discover.

2.2.1.2 Asthma Worsened (Target Variable)

COUNT and ROUND SQL functions revealed a severe class imbalance. Non-worsening conditions are overwhelmingly higher (86.34%) than worsening conditions (13.66%).

```
1  SELECT
2      asthma_worsened,
3      COUNT(*) AS frequency,
4      ROUND(COUNT(*) * 100 / SUM(COUNT(*)) OVER (), 3) AS proportion
5  FROM patients
6  GROUP BY asthma_worsened;
```

Fig 3: SQL code which demonstrates target class imbalance

Status	Count	Percentage
Yes (1)	1,366	13.66%
No (0)	8634	86.34%

Table 2: Asthma deterioration categorisation

2.2.2 Clinical Measurements

2.2.2.1 Blood pressure measurements

Blood pressure measurements show complete data across all 10000 records with no missing values.

Measurement	Count	Mean (/mmHg)	Median (/mmHg)	Min (/mmHg)	Max (/mmHg)	Missing Values
Systolic BP1 (sys_val1)	10000	122.3	122.0	85	150	0
Diastolic BP1 (dias_val1)	10000	78.2	80	50	90	0
Systolic BP2 (sys_val2)	10000	122.5	123.0	85	150	0
Diastolic BP2 (dias_val2)	10000	78.2	80.0	50	90	0

Table 3: Blood pressure readings in dataset

2.2.2.2 Cholesterol Measurements

In the table, a negative minimum HDL value (-0.5) is identified which indicates an entry error and a clear outlier. The higher number of missing values necessitates filling in the missing data or simply removing the variables entirely.

Measurement	Count	Mean (mmol/L)	Median (mmol/L)	Min (mmol/L)	Max (mmol/L)	Missing Values
Total Cholesterol 1 (Chltot_Val1)	5307	6.5	6.5	3.5	9.6	4693
Total Cholesterol 2 (Chltot_Val2)	5307	6.5	6.5	0.5	12.6	4693
HDL Cholesterol 1 (Chlhdl_Val1)	3723	1.3	1.3	0.5	2.0	6277
HDL Cholesterol 2 (Chlhdl_Val2)	3723	1.3	1.2	-0.5	3.0	6277

Table 4: Summary of cholesterol measurements

2.2.2.3 Distance to Motorway

MWAYDIST_KM1 captures 89.8% of patients (10.2% missing). The COUNT function revealed the highest number of patients who lived between 1.4 and 1.5km from the motorways. A right skewed distribution (mean 6.57km > median 4.54km) showed most patients lived closer to the motorways, with a few outliers living further away.

```
1  select
2  mwaydist_km1,
3  COUNT(mwaydist_km1) AS MWAYCOUNT
4  from patients
5  GROUP BY mwaydist_km1
6  ORDER BY MWAYCOUNT DESC
```

Fig 4: SQL code for MWAYDIST_KM1 column inspection

Measurement	Count	% of Total	Mean (km)	Median	Min	Max	Missing Values
Distance (km)	8980	89.8%	6.57	4.54	0.1	29.98	1020

Table 5: A summary of patient proximity to motorways

2.2.3 Clinical SNOMED Codes

A summary of all SNOMED codes are presented below:

Blood Pressure codes: sys_code1/2 (271649006) and dias_code1/2 (271650006), consistent across all 10000 records

Cholesterol codes: chltot_val1/2 (853681000000104) for 7613 records (2387 missing) and chlhdl_val1/2 (1005681000000107) for 9366 records (634 missing)

Medical History:

1. Diabetes History (diab_code1): 1460 patients (14.60% of column entries), 8540 missing

SNOMED Codes	Patient Count
73211009	375
111552007	366
237599002	363
44054006	356

Table 6: Distinct SNOMED codes for Diabetes

2. Smoking Status (smok_code1): 1604 patients (16.04% of column entries), 8396 missing

SNOMED Codes	Patient Count
230056004	299
230057008	270
266919005	267
230058003	267
8392000	253
266920004	248

Table 7: Distinct SNOMED codes for Smoking Status

3. Cardiovascular Disease Medication (cvd_rx_code1): 2349 patients (23.49% of column entries)

SNOMED Codes	Patient Count
372729009	506
6131004	255
11560009	242
1040010010001002	237
11000132102	232
11000129103	231
372727001	217
111708003	215
324121000000109	214

Table 8: Distinct SNOMED codes for Cardiovascular Disease Medication

4. Family History of Asthma: 1146 patients (11.46% of column entries)
- SNOMED Code: 160357008 (100% of column entries)

2.2.4 Date Ranges

From the table below, most clinical measurements were recorded in 2024-2025, with medical history variables recorded from 2023. Proximity to motorway data goes back to January 2021, which could suggest that the data might have been collected in retrospect.

Variable Group	Earliest Date	Latest Date	Missing Dates (Count)
Blood Pressure (sys_val1)	2024-06-22	2025-12-12	0
Blood Pressure (sys_val2)	2024-06-20	2025-12-03	0
Total Cholesterol (chltot_val1)	2024-12-17	2025-12-10	2387
Total Cholesterol (chltot_val2)	2024-12-19	2025-12-12	2387
HDL Cholesterol (chlhdl_val1)	2024-12-17	2025-12-07	637
HDL Cholesterol (chlhdl_val2)	2024-12-22	2025-12-12	637
Medical History (diab_code1, smok_code1, cvdrx_code1, fh_asthma_code1)	2023-12-23	2025-12-12	33441 (in total)
Motorway Distance	2021-01-07	2025-12-12	1020

Table 9: Table of data entry dates for patient data

3.0 Data Preparation (SQL)

A comprehensive SQL script was created to transform the raw dataset into a model-ready format suitable for asthma worsening predictions. It addressed data quality issues through cleaning, standardization, dealing with outliers and feature engineering.

3.1 Addressing Data Quality Issues

Although no null values existed, several columns contained missing or placeholder values and were encoded using the non-NULL representations below:

Issue Type	Column(s)	Action
Invalid Birth years	YEAR_OF_BIRTH	Set C_AGE = NULL, flag c_bad_yob = 1
Negative Distances	MWAYDIST_KM1	Set C_MWAY_EXPOSURE = NULL
BP Outliers	SYS_VAL1/DIAS_VAL1	Flag but retain (c_sys_outlier, c_dias_outlier)
Physiological errors	SYS < DIAS	Flag c_bp_pair_invalid = 1

Table 10: Addressing data quality issues

3.2 SNOMED Code Conversion

Multiple code variables were standardised to interpretable categories. The family history asthma code, for example, was corrected from 160357008 to 160377001 and mapped to a binary category (1 for positive history and 0 for negative history). Smoking status had 6 distinct SNOMED codes, with ambiguous entries set to null.

Variables	Input	Output
Gender (Gender)	446141000124107/446151000124109	M/F (U=cannot determine)
Family History of Asthma (fh_asthma_code1)	160377001 (corrected)/Other	1 or 0
Smoking Status (smok_code1)	230056004, 230057008, 230058003, 266920004	1 = smoker
Diabetes (diab_code1)	44054006, 111552007, 237599002, 73211009	1 = diabetic

Table 11: Converted SNOMED codes

3.2 Feature Engineering

The following features were selected and engineered to capture demographic, environmental and clinical risk factors:

Feature/Variable Name (Cleaned)	Data Type	Calculation logic	Rationale
C_AGE	Continuous	Current year - YEAR_OF_BIRTH	Age could correlate with asthma deterioration
C_MWAY_HIGH_EXPOS	Binary	1 if distance <= 0.5km	Proxy for high air pollution exposure
C_FH_ASTHMA	Binary	1 if code = 160377001, '-' if no code/family history	Genetic predisposition
C_SMOKER	Binary/Null	1 = current , 0= never, Null = unknown	Smoking could cause increased mucus production exacerbating respiratory disease
C_PP	Continuous	SYS_VAL1 - DIAS_VAL1	Pulse pressure
C_HTN	Binary	1 if SYS >= 140 or DIAS >= 90	Hypertension indicator
C_DIABETES	Binary	1 if Diabetes	Diabetes flag
C_CVDRX	Binary	1 if CV medication	Cardiovascular treatment
GENDER	Categorical	N/A	Demographic indicator

Table 12: Engineered features

The features made up demographic (Age, Gender), environmental (pollution exposure), genetic (family history), clinical (blood pressure metrics) and behavioural factors (smoking status). However it is important to determine which features could have a greater predictive power over asthma deterioration (target). This necessitates a thorough exploratory analysis to validate or omit some of the engineered features.

3.2.1 Initial Exploratory Data Analysis (SQL)

3.2.1.1 Asthma Deterioration by Family History

The SQL code below revealed patients with family history showed 29.4% worsening rates compared to lower rates in those without, indicating a genetic impact on deterioration.

```

SELECT
    Fh_Asthma_Code1 AS Family_History,
    COUNT(*) AS Total_Patients,
    SUM(CASE WHEN Asthma_Worsened = 1 THEN 1 ELSE 0 END) AS Worsened_Count,
    ROUND((SUM(CASE WHEN Asthma_Worsened = 1 THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS Worsening_Rate_Percent
FROM
    Patients
GROUP BY
    Fh_Asthma_Code1
ORDER BY
    Family_History;

```

Fig 5: Asthma Deterioration by Family History

FAMILY_HISTORY	TOTAL_PATIENTS	WORSENEO_COUNT	WORSENING_RATE_PERCENT
160357008	1146	337	29.41
-	8854	1029	11.62

Table 13: The relationship between Family History and Asthma Deterioration

3.2.1.2 Asthma Deterioration by Smoking Status

The table below shows a wide variation in asthma worsening rates across SNOMED codes for smoking status (NHS Digital, 2025). Current smoker codes (23#### and 266920004 (trivial smoker) codes), had 26% - 30% worsening rates. Never smokers (266919005) showed 23.22%, and trivial or light smokers showed. Non-smokers (27.67%) aligned more with current light cigarette smokers with the 230056004 SNOMED code.

```

SELECT
    Smok_Code1 AS Smoking_Status,
    COUNT(*) AS Total_Patients,
    SUM(CASE WHEN Asthma_Worsened = 1 THEN 1 ELSE 0 END) AS Worsened_Count,
    ROUND((SUM(CASE WHEN Asthma_Worsened = 1 THEN 1 ELSE 0 END) * 100.0 / COUNT(*)), 2) AS Worsening_Rate_Percent
FROM
    patients
GROUP BY
    Smok_Code1
ORDER BY
    Smoking_Status;

```

Fig 6: Asthma Deterioration by Smoking Status

SMOKING_STATUS	TOTAL_PATIENTS	WORSENEO_COUNT	WORSENING_RATE_PERCENT
8392000	253	70	27.67
230056004	299	83	27.76
230057008	270	81	30
230058003	267	70	26.22
266919005	267	62	23.22
266920004	248	59	23.79
-	8396	941	11.21

Table 14: The worsening rates of Asthma by patient smoking status

Based SQL EDAs, features such as GENDER, AGE, SMOKER, FAMILY_HISTORY_ASTHMA, MOTORWAY_DISTANCE_KM,

HIGH_POLLUTION_EXPOSURE and ASTHMA_WORSENER, are selected for their direct relevance to asthma deterioration. These encapsulated demographic, genetic and environmental risk factors. Diabetes and cardiovascular medication features had little to no predictive power and were dropped. Based on the above EDA, a structured CTE pipeline was implemented to extract, filter and validate features into a clean model ready dataset.

3.3 CTE-Based Modelling View

Common Table Expressions (CTEs) were crucial in enhancing the data quality of this dataset in three stages:

CTE Stages	Rationale	Transformations
<i>base_raw</i>	Extract cleaned and relevant features	Selects appropriate C_* variables from patients table
<i>filtered</i>	Apply final quality filters	Excludes c_bad_yob=1, NULL age/target (n=284 removed)
<i>final_features</i>	Rename for ML model training	Renames C_AGE→age, C_GENDER→gender, etc.

Table 15: The three-stage CTE process

The *base_raw* CTE extracted a subset of features, focusing on cleaned variables and derived columns relevant to later analysis. Extracting these columns enabled the creation of a base dataset for further processing, without impacting the original dataset. The *filtered* CTE referenced *base_raw* to apply filters to ensure data quality by removing incomplete and invalid records, ensuring suitability for modelling. Finally, The *final_features* CTE referenced the filtered CTE to rename the columns for clarity for machine learning. The final, refined result could now be queried to obtain a clean feature dataset for model training. Details of the script can be found in the appendix.

```

filtered AS (
    SELECT *
    FROM base_raw
    WHERE c_bad_yob = 0 -- realistic age
        AND c_age IS NOT NULL
        AND c_asthma_worsened IS NOT NULL
),

final_features AS (
    SELECT
        Patient_ID,
        c_gender          AS gender,
        c_age             AS age,
        c_smoker          AS smoker,
        c_fh_asthma       AS family_history_asthma,
        c_mway_exposure   AS motorway_distance_km,
        c_mway_high_expos AS high_pollution_exposure,
        c_asthma_worsened AS asthma_worsened
    FROM filtered
)

SELECT *
FROM final_features;

-----

-- 4. QUICK CHECK: SAMPLE FROM FEATURE VIEW
-----

SELECT *
FROM asthma_model_features
FETCH FIRST 20 ROWS ONLY;

-----

-- END OF SCRIPT
-----

```

Fig 7: SQL CTE modelling view

PATIENT_ID	GENDER	AGE	SMOKER	FAMILY_HISTORY_ASTHMA	MOTORWAY_DISTANCE_KM	HIGH_POLLUTION_EXPOSURE	ASTHMA_WORSENE
PL7FLSUSIM	F	3	-	0	8.09	0	0
RYDPHSD50C	M	15	-	0	3.16	0	0
W20NN9C2R7	F	8	-	0	3.55	0	0
OXISYUKPJX	U	1	-	0	6.77	0	0
5VJMWOTLUZ	U	15	-	0	4.93	0	0
9TGVS2E03W	F	10	-	0	-	0	0

Table 16: Snapshot representing final view of dataset prepared for model training

4.0 Modelling

4.1 Final Exploratory Data Analysis (MATLAB)

The engineered dataset was uploaded into MATLAB for modelling. Univariate and bivariate EDAs performed identified patterns, outliers and anomalies to validate final feature selection.

4.1.1 Age

Age distribution feature reveals a younger population with median and average ages of 39 and 42 respectively. Implausible outliers (ages 0 and 105) were removed.

```
>> summary(Z.C_AGE);  
  
10000x1 double  
  
NumMissing    0  
Min            0  
Median        39  
Max           105  
Mean          42.2140  
Std           26.4970
```

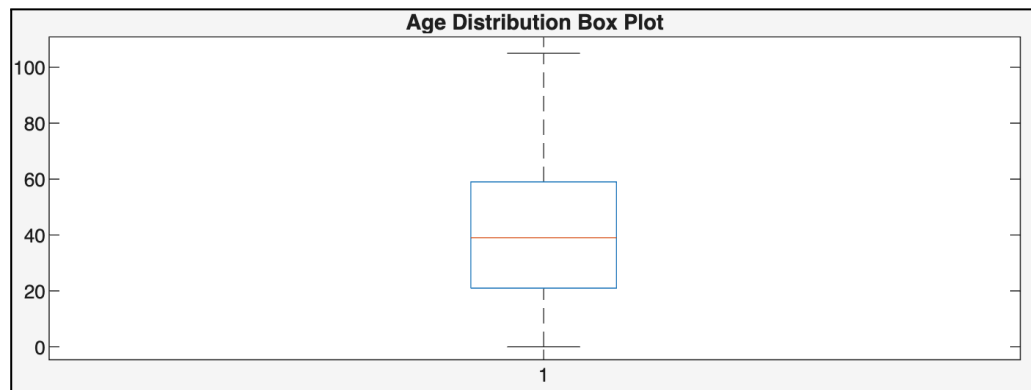


Fig 8: Age Distribution

4.1.2 Motorway Proximity

Most sampled patients live within 0 to 15km from the motorways, suggesting that motorway proximity is a strong factor in asthma symptom worsening.

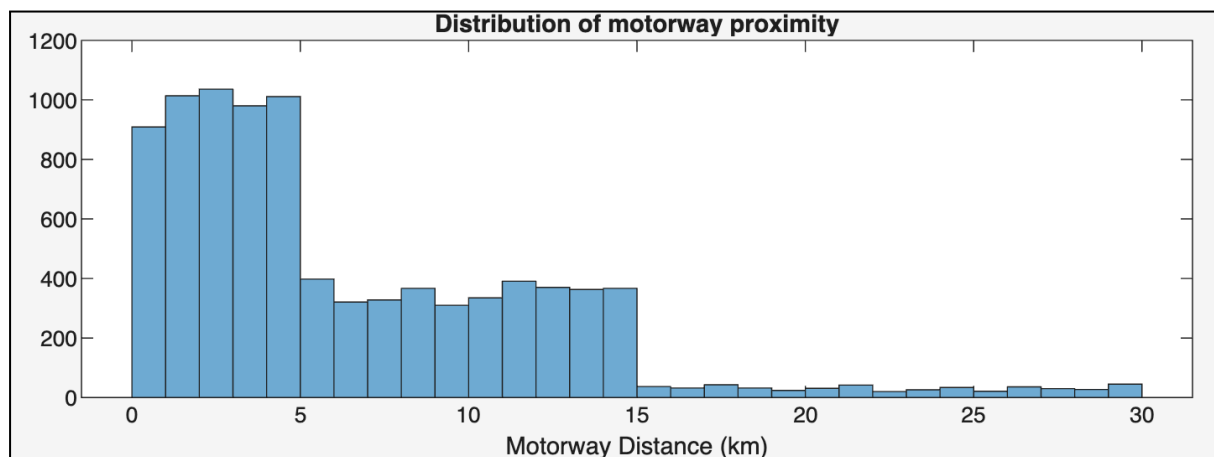


Fig 9: Patient proximity to motorways

4.1.3 Target Distribution

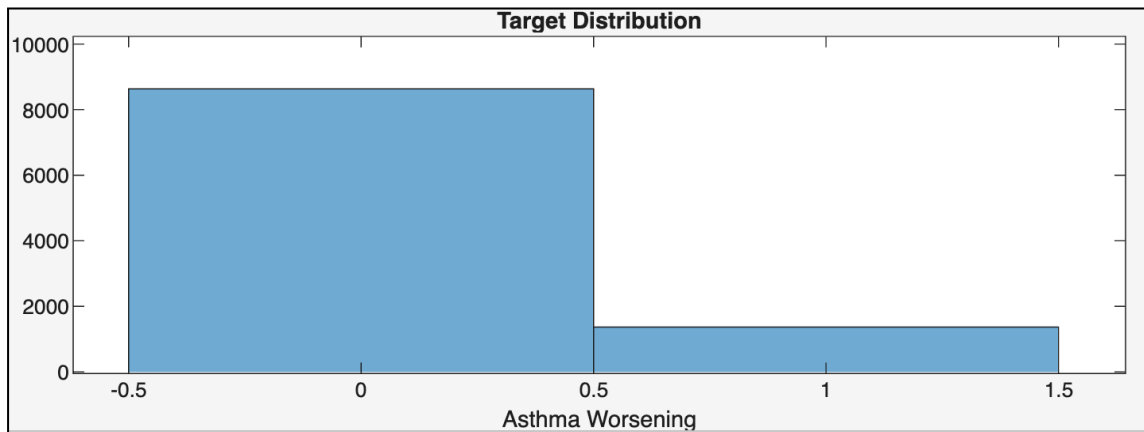


Fig 10: Asthma Worsening cases

ASTHMA_WORSENING shows a clear class imbalance favouring non-worsening over worsening conditions.

4.1.4 Age v Asthma Worsening

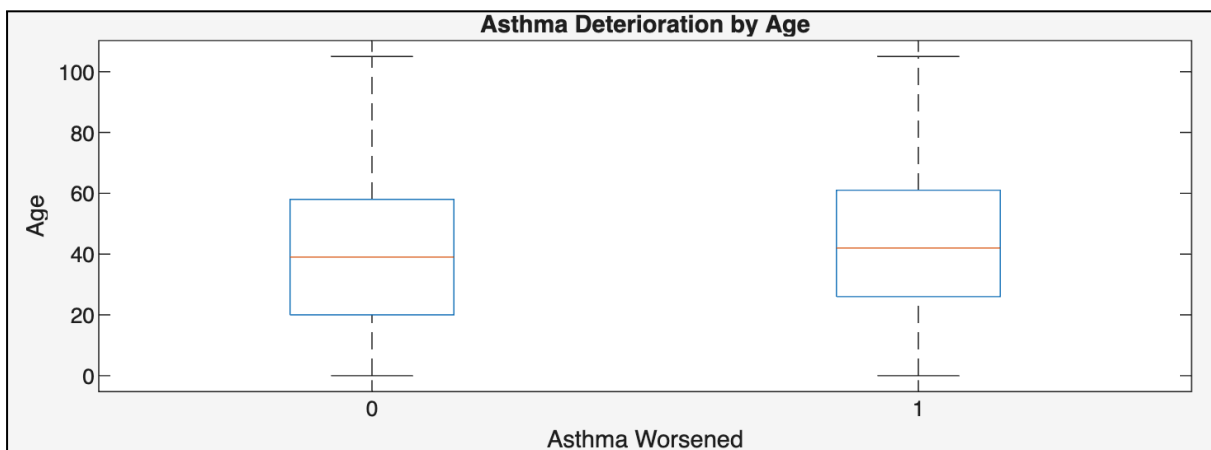


Fig 11: Patient Asthma deterioration by Age

Box plot analysis reveals slightly similar median ages between worsening and non-worsening conditions. That demonstrates that asthma deterioration occurs across comparable age ranges, around adult midlife.

4.1.5 Motorway Proximity v Asthma Worsening

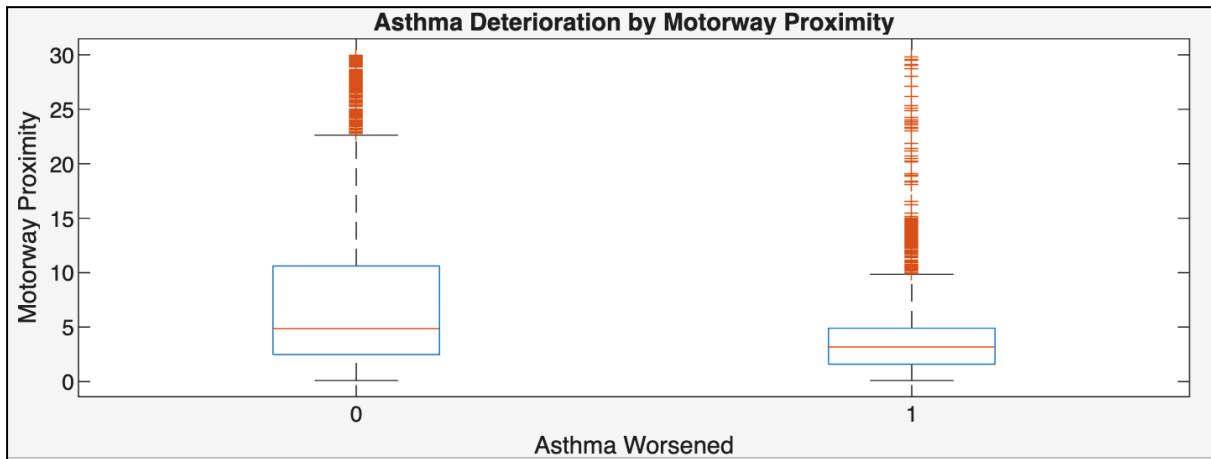


Fig 12: Motorway proximity by Asthma condition metrics

Clear correlation exists between closer motorway proximity and asthma symptoms worsening. While median distances are similar, the lack of worsened cases beyond 15km implies a potential threshold effect. Living beyond this point may mitigate pollution exposure risks. This aligns with epidemiological studies on air quality and respiratory health, where proximity to high-traffic areas increases exposure to pollutants that can exacerbate asthma (Hahn et al., 2008).

4.2 Feature Categorisation and Model Selection

ASTHMA_WORSENERD remained the target feature while GENDER, SMOKER, FAMILY_HISTORY, HIGH_POPULATION_EXPOSURE, were changed to categorical using the MATLAB categorical() function. However, AGE and MOTORWAY_DISTANCE remained numeric.

4.3 Train/Test Split

Train/Test split used cvpartition() function, with 0.3 Holdout, to assess model predictive performance. 70% of the dataset was allocated for training the model, with 30% reserved for testing.

4.4 Model training and selection

Three (3) classification models were trained and evaluated: The Decision Tree Classifier, Logistic Regression and Random Forest Models. Details on the dataset upload, EDAs and training and evaluations of the models appear in the appendix.

4.4.1 Decision Tree Classifier

The Decision Tree model was chosen for interpretability and ability to handle mixed data types typical of healthcare datasets. It serves as an excellent baseline model for comparison (Hahn et al., 2008). The model was trained using *fitctree()* with *MinNumSplits* to reduce branches.

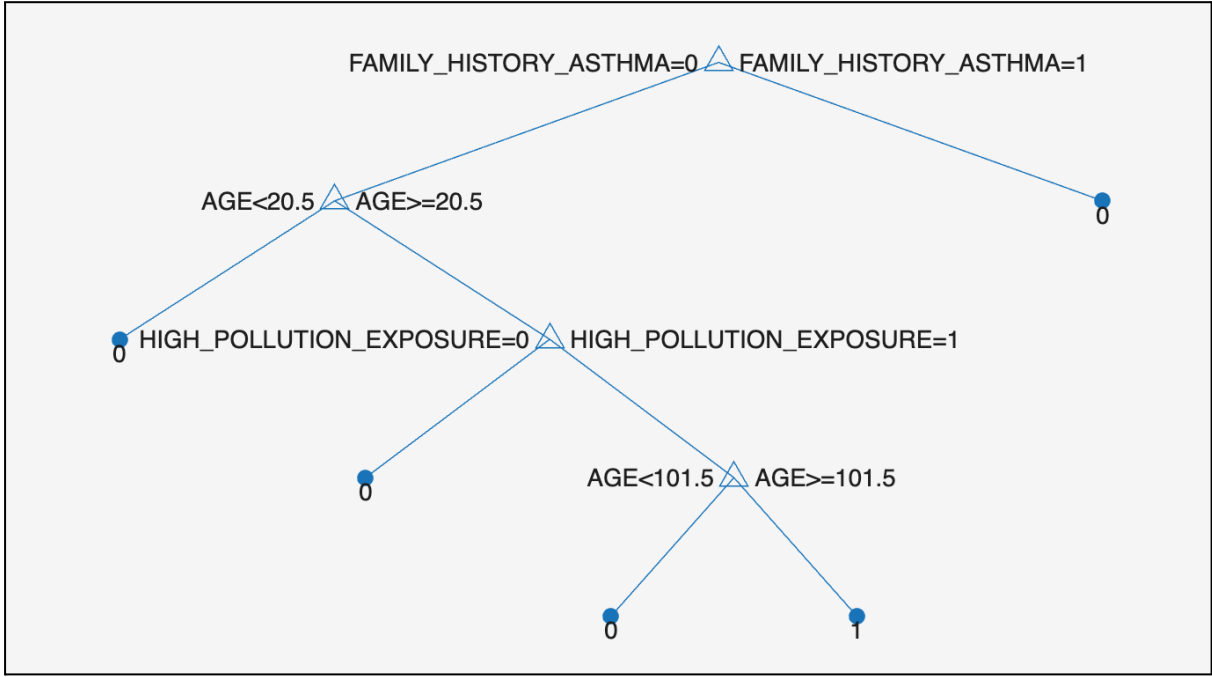


Fig 13: Decision Tree Classifier Model

FAMILY_HISTORY emerged as the most influential predictor, followed by age and high pollution exposure. However most leaves predict non-worsening (0) often, compared to worsening (1), highlighting weak class discrimination.

4.4.1.1 Findings

Its confusion matrix provides insights on accuracy, precision and recall/sensitivity. These are crucial metrics, which indicates the model’s capacity to make mistakes.

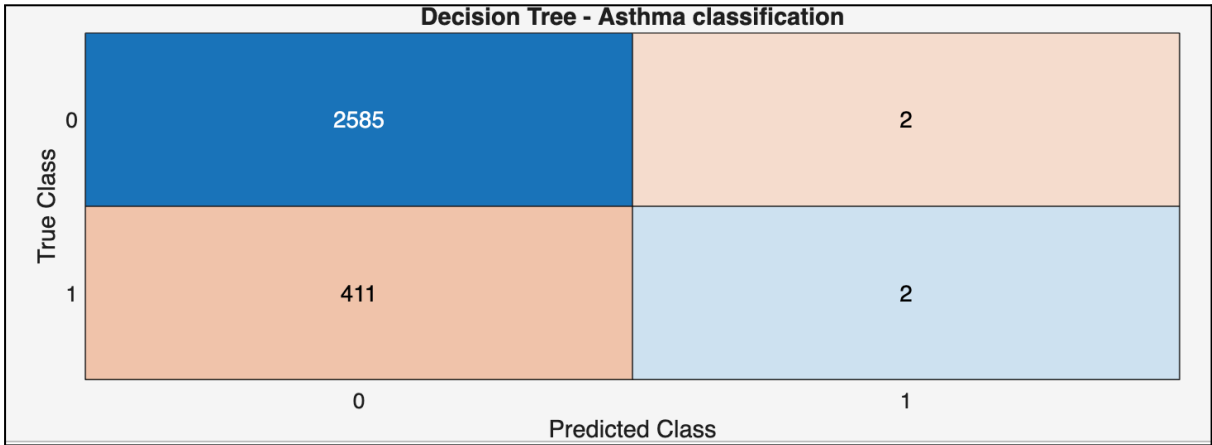


Fig 14: Confusion Matrix - Decision Tree Classifier

4.4.1.1.1 Accuracy

$$Accuracy = \frac{TN+TP}{(TN+FP+TN+TP)} = \frac{2+2585}{3000} = 0.8623 = 86.23\%$$

This high accuracy is misleading due to class imbalance. The model prioritises the majority class but disregards the minority.

4.4.1.1.2 Precision

$$Precision = \frac{TP}{(TP+FP)} = \frac{2}{2+2} = 0.500 = 50\%$$

50% precision means a 50% false alarm rate, which is acceptable and could be mitigated.

4.4.1.1.3 Recall/Sensitivity

$$Recall = \frac{TP}{(TP+FN)} = \frac{2}{2+411} = 0.0048 = 0.48\%$$

The model missed 411 out of 413 predicted cases rendering it unsuitable for deployment.

4.4.1.1.4 ROC Curve (AUC)

The ROC curve evaluates a binary classification performance, by plotting the sensitivity (True positive rate) against False positive rate.

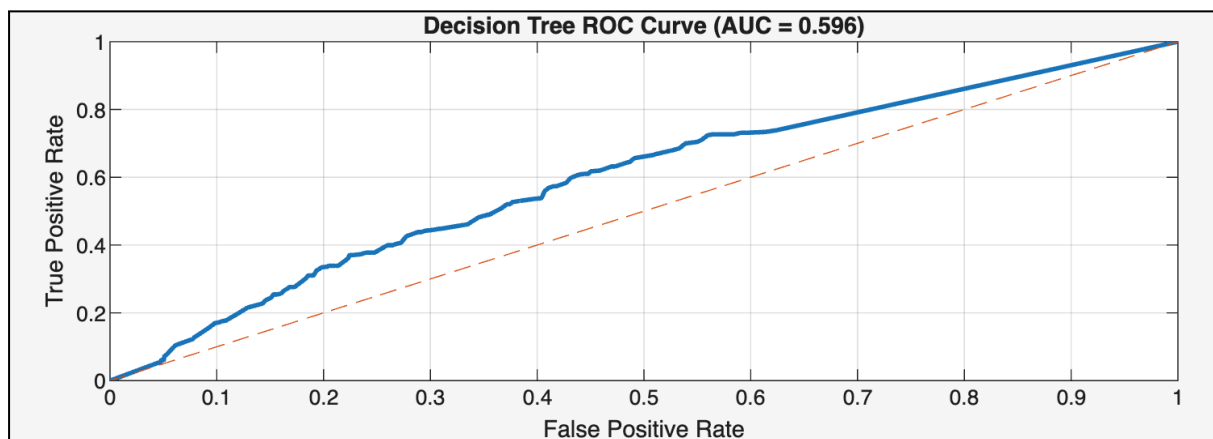


Fig 15: The ROC/AUC curve of the Decision Tree Classifier

The AUC value of 0.595 indicates a weak predictive quality, marginally better than making random predictions (AUC = 0.5). Limited features, and their binary nature constrain the model's ability to separate worsening from non-worsening cases.

4.4.2 Logistic Regression Model

Given the binary target (ASTHMA_WORSENING), this model ascertains the probability of each binary class. It serves as a good benchmark, widely used in healthcare related studies.

4.4.2.1 Findings

Based on the engineered dataset, the outcome of the trained model is outlined on the confusion matrix below:

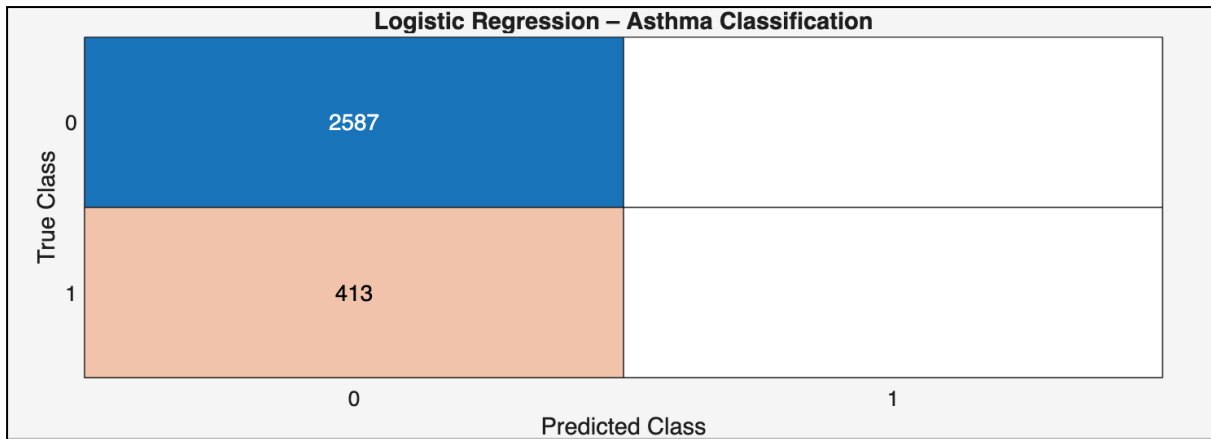


Fig 16: Confusion matrix of the Logistic Regression model

4.4.2.1.1 Accuracy

$$Accuracy = \frac{TN+TP}{(TN+FP+TN+TP)} = \frac{2587}{3000} = 0.8623 = 86.23\%$$

This 86.23% reflects majority class prediction only, demonstrating model naivety regarding the minority class.

4.4.2.1.2 Precision

$$Precision = \frac{TP}{(TP+FP)} = \frac{0}{0+0} = 0\%$$

No true or false positives exist, making precision unattainable. No resource overallocation would occur, but the model lacks positive predictive capability.

4.4.2.1.3 Recall/Sensitivity

$$Recall = \frac{TP}{(TP+FN)} = \frac{0}{0+413} = 0\%$$

All 413 worsening cases were incorrectly predicted as non worsening. This presents an alarming error, highlighting poor model predictive capability.

4.4.2.1.4 ROC curve (AUC)

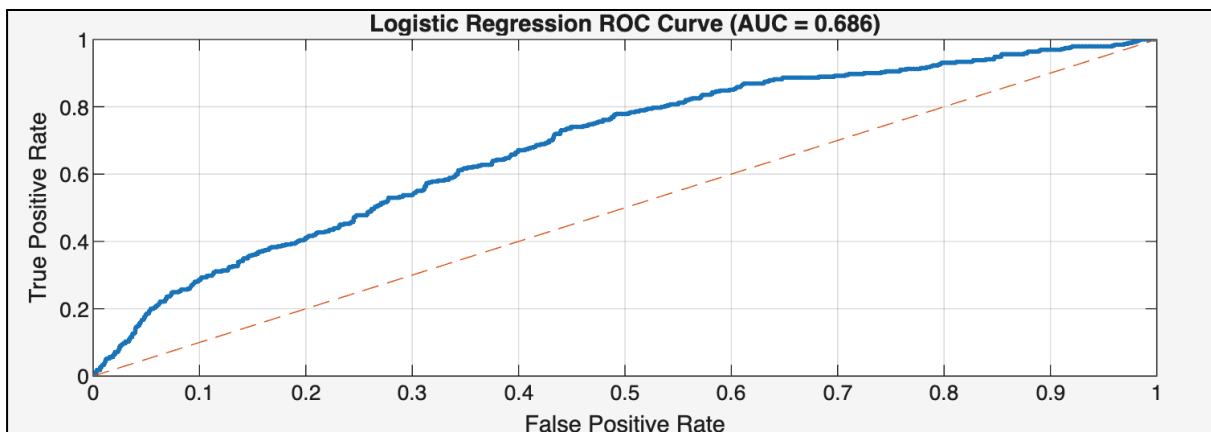


Fig 17: ROC curve of the Logistic Regression Model

AUC value of 0.686 outperforms the Decision Tree model, indicating improved performance. The model can rank worsening cases higher than non-worsening ones, even without acting on it at the current threshold. This demonstrates meaningful feature learning, even though the value is below ideal (>0.8).

4.4.3 Random Forest Model

Similar accuracy and AUC values across models warranted a third comparison. The Random Forest Model was selected to investigate if it would yield favourable outcomes for sensitivity and higher AUC values.

4.4.3.1 Findings

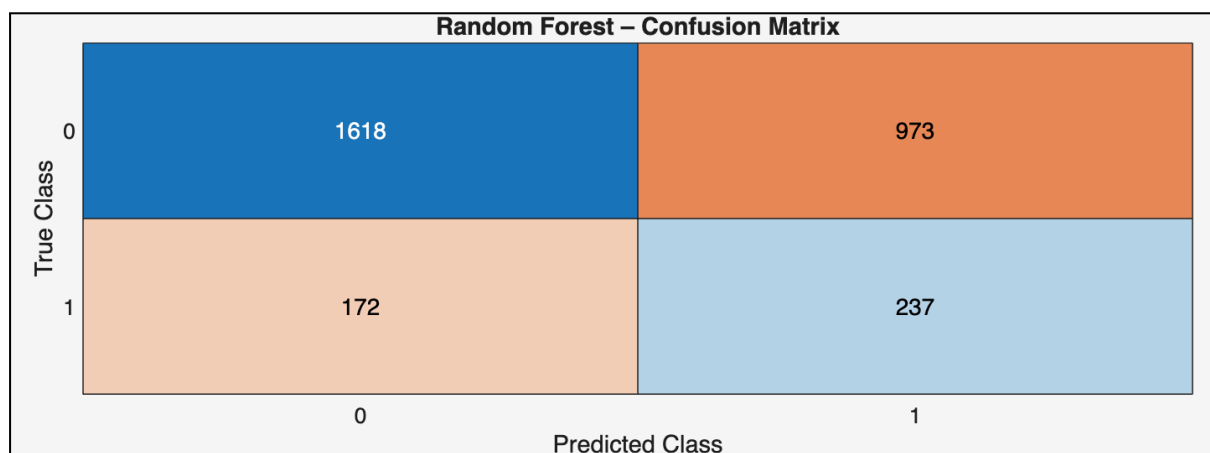


Fig 18: Confusion Matrix for Random Forest Model

4.4.3.1.1 Accuracy

$$Accuracy = \frac{TN+TP}{(TN+FP+TN+TP)} = \frac{1618+237}{3000} = 0.618 = 61.8\%$$

Lower accuracy (61.8%) reflects trade-offs with its sensitivity metrics. Its ensemble characteristics handle class imbalance effectively.

4.4.3.1.2 Precision

$$Precision = \frac{TP}{(TP+FP)} = \frac{237}{237+973} = 0.196 = 19.6\%$$

Precision (19.6%) falls between DT's 50% and LR's 0%. High false positives rates are acceptable in this use case, but would be problematic in domains like Finance.

4.4.3.1.3 Recall/sensitivity

$$Recall = \frac{TP}{(TP+FN)} = \frac{237}{237+172} = 0.579 = 57.9\%$$

237 of 409 patients were correctly predicted. This highest recall/sensitivity demonstrates strong model predictability suitable for real world high-stakes predictions in this case.

4.4.3.1.4 ROC curve (AUC)

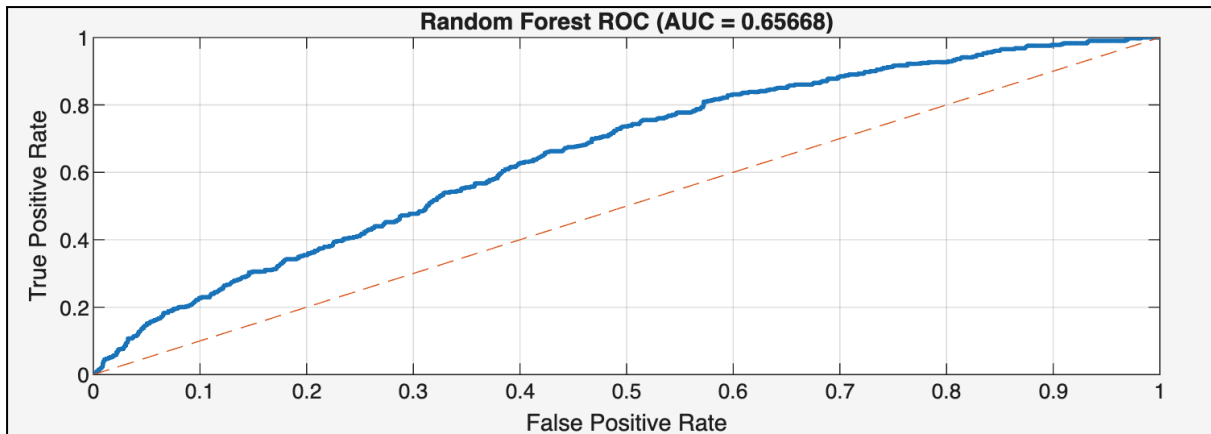


Fig 19: ROC/AUC for Random Forest Model

AUC value of 0.657, outperforms DT but shows inadequate class distinction capability.

In summary, the Random Forest model demonstrated superior model sensitivity. Given the importance of detecting worsening cases to enable interventions, model sensitivity was prioritised, where the Random Forest model excelled.

5.0 Evaluation

5.1 Model Performance Comparisons

The Random Forest model remained the best choice for predicting worsening asthma cases. In this medical context, class imbalance toward non-worsening cases made sensitivity (recall) the primary criteria of evaluation. Minimising false negatives was crucial, since undetected worsening symptoms could delay clinical interventions and cause serious patient harm.

The Random Forest’s sensitivity of 57.9% substantially outperformed both the Decision Tree (0.48%) and Logistic Regression (0%), which were essentially ineffective at identifying worsening cases. Although Decision Tree and Logistic Regression models recorded deceptively high accuracies (86.23%), the metric only reflected correct prediction of the majority non worsening class, offering little clinical value. The Random Forest’s lower accuracy (61.8%) but higher precision (19.6%) represented a necessary trade-off: false positives can be managed through follow-up monitoring without the severe consequences of missed deterioration cases.

While the Logistic Regression model had slightly higher AUC (0.686 vs the RF’s 0.657), the overall poor values across the models revealed inadequate class distribution capability. Additionally, its sensitivity (0%) rendered it clinically unusable.

Summary of Model Performance Metrics

Metric	Decision Tree	Logistic Regression	Random Forest Model
Accuracy	86.23%	86.23%	61.8%
Precision	50%	0% (undefined)	19.6%
Sensitivity	0.48%	0%	57.9%
AUC	0.595	0.686	0.657

Table 17: Summary of model performance

5.2 Feature Contribution

FAMILY_HISTORY_ASTHMA and MOTORWAY_DISTANCE_KM contributed most significantly to the model's predictive performance. Initial correlation analyses (EDA) indicated strong correlations between these variables and the target, suggesting direct relevance in identifying asthma deterioration. Both Decision Tree and Random Forest Models confirmed tree-based feature importance for these variables in detecting worsening cases.

MOTORWAY_DISTANCE_KM is a strong predictor because patients living near major motorways face higher nitrogen dioxide exposure, which are known asthma triggers. FAMILY_HISTORY_ASTHMA indicated genetic proclivities, while age ranges reflected disease progression in patients over time. SMOKER, though a well established risk factor, showed a limited contribution due to incomplete recording, or underrepresentation of smoking behaviour across patient groups. It furthermore warrants further investigation in larger datasets, where indulgence or exposure to smoking could be better put in context.

5.3 Limitations and Methodical Errors

Results were limited by several constraints. Class imbalance challenged all three model's ability to learn worsening case patterns. The synthetic data's minimal features did not fully encapsulate real-world datasets complexity, limiting clinical applicability. Also, predicting Asthma worsening symptoms depend on multiple interrelated features. More features improve model sensitivity and its predictive capability.

5.4 Model Usefulness and Recommendations

The Random Forest Model has improvement potential through techniques like Threshold Tuning or Oversampling and resampling methods which generate synthetic samples for the minority class enabling a balance of the dataset to improve model sensitivity. For example, applying Adaptive Synthetic Sampling (ADASYN) enables Random Forest to reach even higher recalls, ensuring that cases of clinical interest are accurately identified (Kushi et al., 2024).

6.0 Deployment recommendations

6.1 GP Practice Limitations

Models like this require large and diverse datasets to detect uncountable asthma symptom causes (Greenhalgh et al., 2019). Deploying the Random Forest model at the GP practice level risks data quality issues, including misaligned SNOMED codes and entry inconsistencies. Also, localised GP practices lack varied socio-economic, environmental indicators needed for optimal performance. The risk of a skew in demographics could also exist, potentially biasing the outcome of the model's prediction.

6.2 High Volume Data Challenges

National level deployment supports the model's capabilities since it provides huge volumes of data and features to boost its predictive capabilities (Khudur et al., 2025). However, integrating distributed data across practices presents timing, synchronization, governance, privacy and infrastructural issues. Effective national rollout then requires appropriate architecture to mitigate the issues, while maintaining historical patient data accuracy.

6.3 Architecture, Guidelines and Recommendation

Cloud-based systems, Microservices and Kappa/Lambda architectures can manage big data effectively (Behera and Behera et al., 2017). They do this through scaling to manage and process petabytes of clinical or IoT data without performance degradation (Singh et al., 2021).

6.3.1 Cloud-Based Architecture

Cloud-based systems provide centrally managed, cost-effective platforms via secure NHS approved cloud environments, addressing big data's 7 V's. In context, they efficiently process real time sensor/wearable data (e.g., AWS for smart inhalers) while supporting historical population analytics, without microservices' fault propagation risks (Hastie, Tibshirani and Friedman, 2009). In comparison, Microservices' database-per-service pattern where each service maintains its own data store (Alshuqayran, Ali and Evans, 2016), fragments data, hindering unified data access required for population-scale models. Making standalone use of them is inadvisable as a result.

6.3.2 Kappa and Lambda Architectures

The Kappa architecture prioritises real-time stream processing but lacks historical auditing for safety-critical NHS population health analytics (Xie et al., 2025). Lambda's batch layer, in

comparison, prioritises historical data, given asthma worsening's dependence on historical and environmental patterns. Its speed layer enables real-time data collection. The servicing layer supports querying for model training. This batch layer makes Lambda superior for large data management.

6.3.3 Regulatory Guidelines

NHS deployment requires UK GDPR Article 9, the Data Security and Protection Toolkit (DSPT), and NHS Digital's standards compliance (Data Protection Act 2018, c.12). Cloud designs' governance and scalability suit safety critical, population scale analytics.

Architectures must demonstrate DCBO129 compliance, favouring centralised platforms where security perimeters and clinical safety controls can be uniformly applied and audited (Khudur et al., 2025).

6.3.4 Recommended Architecture

Based on real world data mining practices, the **Lambda architecture** is the best choice for the following reasons:

1. Its Hybrid capability: It delivers accurate historical patient data through its batch layer, while its speed layer allows real time processing to handle alerts.
2. NHS specific scalability and compliance: Lambda handles fragmented data from GPs, while meeting GDPR and DSTP compliance. According to Compare the Cloud (2025), the NHS Federated Platform (FDP) uses cloud native architectures with warehousing (Snowflake) and streaming (Kafka) to secure the sharing of over 50million records.
3. Data mining best practices where experts recommend it for mixed workloads, having the freedom to make latency secondary to patient data accuracy.

The Lambda architecture deployed on a cloud platform is the most ideal choice given our use case. It balances the need for a robust historical analysis (identifying asthma risk patterns over time) with real time processing. It does this while ensuring scalability and compliance within a national NHS context.

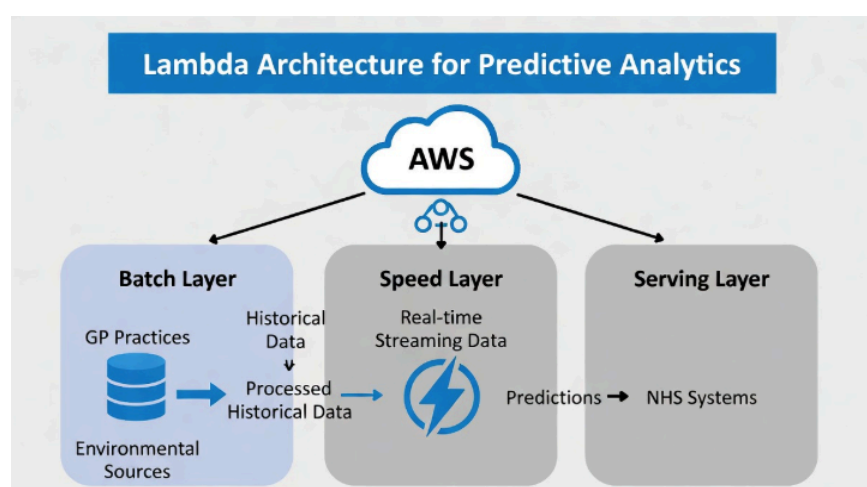


Fig 20: Illustration of the Lambda architecture on a cloud platform

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17. Copilot to tighten text sentences and to provide an overarching understanding of certain concepts

8.0 Appendix

8.1 SQL Script

```
-----  
-- PATIENT DATA: CLEANING + FEATURE ENGINEERING (WITH CTEs)  
-- Table: PATIENTS  
-- CI7524 Big Data & Data Mining - Anthony L. Anumel  
-- 1. Add derived columns (safe / idempotent)  
-- 2. Populate cleaned + feature columns  
-- 3. Create a CTE-based modelling view  
-----  
  
-----  
-- (OPTIONAL) BACKUP ORIGINAL TABLE
```

```

-----
-- CREATE TABLE patients_raw AS SELECT * FROM patients;
-- (Uncomment and run once if you want a snapshot)

-----

-- 1. ADD DERIVED COLUMNS (SAFE / IDEMPOTENT)
-----

DECLARE
    col_count INTEGER;

-- Helper: add column only if it does NOT already exist
PROCEDURE add_column(colname IN VARCHAR2, datatype IN VARCHAR2) IS
BEGIN
    SELECT COUNT(*)
    INTO col_count
    FROM user_tab_columns
    WHERE table_name = 'PATIENTS'
    AND column_name = UPPER(colname);

    IF col_count = 0 THEN
        EXECUTE IMMEDIATE 'ALTER TABLE patients ADD ' || colname || ' ' ||
datatype;
    END IF;
END;

BEGIN
    -- Cleaned demographic + BP features
    add_column('C_GENDER', 'VARCHAR2(1)'); -- cleaned gender: M/F/U
    add_column('C_AGE', 'NUMBER'); -- cleaned age
    add_column('C_SYS_OUTLIER', 'NUMBER(1)'); -- 1 = out-of-range sys_val1
    add_column('C_DIAS_OUTLIER', 'NUMBER(1)'); -- 1 = out-of-range dias_val1
    add_column('C_BP_PAIR_INVALID', 'NUMBER(1)'); -- 1 = sys < dias
    add_column('C_PP', 'NUMBER'); -- pulse pressure

    -- Cholesterol + clinical flags
    add_column('C_CHOL_OUTLIER', 'NUMBER(1)'); -- 1 = suspicious chol
    add_column('C_DIABETES', 'NUMBER(1)'); -- 1 = diabetes dx
    add_column('C_CVDRX', 'NUMBER(1)'); -- 1 = on cv meds
    add_column('C_HTN', 'NUMBER(1)'); -- 1 = hypertensive

    -- Asthma related features
    add_column('C_FH_ASTHMA', 'NUMBER(1)'); -- 1 = family history of asthma
    add_column('C_MWAY_EXPOSURE', 'NUMBER'); -- distance to motorway (km)
    add_column('C_SMOKER', 'NUMBER(1)'); -- 1 = smoker, 0 = non, NULL =
unknown

```

```

add_column('C_MWAY_HIGH_EXPOS', 'NUMBER(1)'); -- 1 = high pollution exposure
add_column('C_ASTHMA_WORSENEDE', 'NUMBER(1)'); -- target variable

-- Data quality flags
add_column('C_BAD_YOB', 'NUMBER(1)'); -- 1 = invalid Year_of_Birth
add_column('C_BAD_BP_DATE1', 'NUMBER(1)'); -- 1 = invalid bp_date1
add_column('C_BAD_CHOL_DATE1', 'NUMBER(1)'); -- 1 = invalid chlhdl_date1
END;
/

-----

-- 2. POPULATE CLEANING + FEATURE COLUMNS
-----

UPDATE patients
SET

-----

-- 2.1 CLEAN GENDER (SNOMED → M/F/U)
-- male_code = '446141000124107'
-- female_code = '446151000124109'
-----

c_gender =
CASE
    WHEN Gender = '446141000124107' THEN 'M'
    WHEN Gender = '446151000124109' THEN 'F'
    ELSE 'U'
END,

-----

-- 2.2 CLEAN FAMILY HISTORY OF ASTHMA
-- family_history_asthma_code = "160377001"
-----

c_fh_asthma =
CASE
    WHEN fh_asthma_code1 = '160377001' THEN 1
    ELSE 0
END,

-----

-- 2.3 CLEAN MOTORWAY EXPOSURE DISTANCE
-----

c_mway_exposure =
CASE
    WHEN mwaydist_kml IS NULL THEN NULL
    WHEN mwaydist_kml < 0 THEN NULL -- invalid distance; no such thing as
a negative distance

```

```

        ELSE mwaydist_kml                                -- preserve float value (in km)
    END,

-----
-- 2.4 CLEAN AGE FROM YEAR_OF_BIRTH
-----

c_age =
    CASE
        WHEN Year_of_Birth IS NULL THEN NULL
        WHEN Year_of_Birth < 1900 THEN NULL
        WHEN Year_of_Birth > EXTRACT(YEAR FROM SYSDATE) THEN NULL
        ELSE EXTRACT(YEAR FROM SYSDATE) - Year_of_Birth
    END,

c_bad_yob =
    CASE
        WHEN Year_of_Birth IS NULL THEN 0
        WHEN Year_of_Birth < 1900 THEN 1
        WHEN Year_of_Birth > EXTRACT(YEAR FROM SYSDATE) THEN 1
        ELSE 0
    END,

-----
-- 2.5 BLOOD PRESSURE VALUE CHECKS & PULSE PRESSURE
-- systolic: 60-260
-- diastolic: 30-160
-----

c_sys_outlier =
    CASE
        WHEN sys_val1 IS NULL THEN 0
        WHEN sys_val1 < 60 OR sys_val1 > 260 THEN 1
        ELSE 0
    END,

c_dias_outlier =
    CASE
        WHEN dias_val1 IS NULL THEN 0
        WHEN dias_val1 < 30 OR dias_val1 > 160 THEN 1
        ELSE 0
    END,

c_bp_pair_invalid =
    CASE
        WHEN sys_val1 IS NOT NULL
            AND dias_val1 IS NOT NULL

```

```

        AND sys_val1 < dias_val1 THEN 1
    ELSE 0
END,

c_pp =
    CASE
        WHEN sys_val1 IS NOT NULL AND dias_val1 IS NOT NULL
            THEN sys_val1 - dias_val1
        ELSE NULL
    END,

-----
-- 2.6 CHOLESTEROL OUTLIER FLAG (1-20 mmol/L plausible)
-----

c_chol_outlier =
    CASE
        WHEN chlhdl_val1 IS NULL THEN 0
        WHEN chlhdl_val1 < 1 OR chlhdl_val1 > 20 THEN 1
        ELSE 0
    END,

-----
-- 2.7 HIGH POLLUTION EXPOSURE FLAG (0.5KM is viable)
-----

c_mway_high_expos =
    CASE
        WHEN mwaydist_km1 IS NOT NULL AND mwaydist_km1 <= 0.5 THEN 1
        ELSE 0
    END,

-----
-- 2.8 ASTHMA WORSENING (TARGET)
-----

c_asthma_worsened =
    CASE
        WHEN asthma_worsened IS NULL THEN 0
        WHEN asthma_worsened = 1 THEN 1
        ELSE 0
    END,

-----
-- 2.9 DIABETES FLAG
-- diabetes_codes = 44054006,111552007,237599002,73211009
-----

c_diabetes =
    CASE

```

```

        WHEN diab_code1 IN ('44054006','111552007','237599002','73211009')
            THEN 1
        ELSE 0
    END,

-----

-- 2.10 SMOKING FLAG
-- all smoking codes:
-- 8392000,230056004,230057008,230058003,266919005,266920004
-- smoker=1:
-- 230056004,230057008,230058003,266920004
-----

c_smoker =
    CASE
        WHEN smok_code1 IN ('230056004','230057008','230058003','266920004')
THEN 1  -- smoker
        WHEN smok_code1 IN ('8392000','266919005') THEN NULL
-- ambiguous
        WHEN smok_code1 IS NULL THEN NULL
-- unknown
        ELSE 0
-- explicit non-smoker (if coded)
    END,

-----

-- 2.11 CARDIOVASCULAR MEDICATION FLAG
-- bp_management_rx_codes:
-- 6131004,372727001,11000132102,11000129103,
-- 1040010010001002,111708003,372729009,
-- 324121000000109,11560009
-----

c_cvdrx =
    CASE
        WHEN cvdrx_code1 IN (
            '6131004','372727001','11000132102','11000129103',
            '1040010010001002','111708003','372729009',
            '324121000000109','11560009'
        ) THEN 1
        ELSE 0
    END,

-----

-- 2.12 HYPERTENSION FLAG
-- c_htn = 1 if sys ≥ 140 OR dias ≥ 90
-----

```

```

c_htn =
    CASE
        WHEN sys_val1 >= 140 OR dias_val1 >= 90 THEN 1
        ELSE 0
    END,

-----

-- 2.13 DATE QUALITY FLAGS
-----

c_bad_bp_date1 =
    CASE
        WHEN bp_date1 IS NULL THEN 0
        WHEN bp_date1 < DATE '1900-01-01' THEN 1
        WHEN bp_date1 > SYSDATE THEN 1
        ELSE 0
    END,

c_bad_chol_date1 =
    CASE
        WHEN chlhdl_date1 IS NULL THEN 0
        WHEN chlhdl_date1 < DATE '1900-01-01' THEN 1
        WHEN chlhdl_date1 > SYSDATE THEN 1
        ELSE 0
    END

;

COMMIT;

-----

-- 3. FEATURE VIEW FOR MODELLING (USING CTE PIPELINE)
-----

-- CTE chain:
--   base_raw      → filtered
--   filtered      → final_features
-----

CREATE OR REPLACE VIEW asthma_model_features AS
WITH base_raw AS (
    -----
    -- Step 1: Start from cleaned columns on PATIENTS
    -----

    SELECT
        Patient_ID,
        c_gender,

```

```

        c_age,
        c_smoker,
        c_fh_asthma,
        c_bad_yob,
        c_mway_exposure,
        c_mway_high_expos,
        c_asthma_worsened
    FROM patients
),

filtered AS (
    SELECT *
    FROM base_raw
    WHERE c_bad_yob = 0 -- realistic age
        AND c_age IS NOT NULL
        AND c_asthma_worsened IS NOT NULL
),

final_features AS (
    SELECT
        Patient_ID,
        c_gender           AS gender,
        c_age              AS age,
        c_smoker           AS smoker,
        c_fh_asthma        AS family_history_asthma,
        c_mway_exposure    AS motorway_distance_km,
        c_mway_high_expos  AS high_pollution_exposure,
        c_asthma_worsened  AS asthma_worsened
    FROM filtered
)

SELECT *
FROM final_features;

-----
-- 4. QUICK CHECK: SAMPLE FROM FEATURE VIEW
-----

SELECT *
FROM asthma_model_features
FETCH FIRST 20 ROWS ONLY;

-----
-- END OF SCRIPT
-----

```

8.2 MATLAB Script

```
>> % load engineered dataset
>> T=readtable("asthma_model_features.csv");
>>
```

```
>> % observe first few rows
>> head(T) % the first few rows
```

PATIENT_ID	GENDER	AGE	SMOKER	FAMILY_HISTORY_ASTHMA	MOTORWAY_DISTANCE_KM	HIGH_POLLUTION_EXPOSURE	ASTHMA_WORSENERD
{'PL7FLSUSIM'}	{'F'}	3	NaN	0	8.09	0	0
{'RYDPHSD50C'}	{'M'}	15	NaN	0	3.16	0	0
{'W20NN9C2R7'}	{'F'}	8	NaN	0	3.55	0	0
{'OXISYUKPJX'}	{'U'}	1	NaN	0	6.77	0	0
{'5VJMWOTLUZ'}	{'U'}	15	NaN	0	4.93	0	0
{'9TGVS2E03W'}	{'F'}	10	NaN	0	NaN	0	0
{'S064UKX1KC'}	{'U'}	14	NaN	0	8.32	0	0
{'B1JSV3KJQQ'}	{'F'}	0	NaN	0	NaN	0	1

```
>> summary(T) % summary of dataset by columns
```

T: 10000×8 table

Variables:

```
PATIENT_ID: cell array of character vectors
GENDER: cell array of character vectors
AGE: double
SMOKER: double
FAMILY_HISTORY_ASTHMA: double
MOTORWAY_DISTANCE_KM: double
HIGH_POLLUTION_EXPOSURE: double
ASTHMA_WORSENERD: double
```

Statistics for applicable variables:

	NumMissing	Min	Median	Max	Mean	Std
PATIENT_ID	0					
GENDER	0					
AGE	0	0	39	105	42.2140	26.4970
SMOKER	8916	1	1	1	1	0
FAMILY_HISTORY_ASTHMA	0	0	0	1	0.1146	0.3186
MOTORWAY_DISTANCE_KM	1020	0.1000	4.5400	29.9800	6.5673	5.7193
HIGH_POLLUTION_EXPOSURE	0	0	0	1	0.0404	0.1969
ASTHMA_WORSENERD	0	0	0	1	0.1366	0.3434

```
>> % exploratory analyses of features
>> figure; histogram(T.AGE);
>> title('Age Distribution');
>>
>> figure; histogram(T.MOTORWAY_DISTANCE_KM);
>> title('Motorway Distance');
>>
>> figure; scatter(T.AGE, T.SMOKER);
>> title('Correlation between age and smoker');
```

```
>> figure; boxplot(T.C_AGE);
>> title('Age Distribution Box Plot');
>>
>> figure; histogram(T.ASTHMA_WORSENERD);
>> title('Target Distribution')
>> xlabel('Target Classes')
>>
>> figure; boxplot(T.C_AGE, T.C_ASTHMA_WORSENERD);
>> title('Asthma Deterioration by Age');
>> xlabel('Asthma Worsened')
>> ylabel('Age')
>>
>> figure; boxplot(T.MWAYDIST_KM1, T.ASTHMA_WORSENERD);
>> title('Asthma Deterioration by Motorway Proximity')
>> xlabel('Asthma Worsened')
>> ylabel('Motorway Proximity')
```

Modelling

```
>> % Modeling
>> Y = T.ASTHMA_WORSENER; % define the target
>> % perform the train and test split
>> X = T(:, {'GENDER', 'FAMILY_HISTORY_ASTHMA', 'HIGH_POLLUTION_EXPOSURE'}); % define predictors, which are a subset of useful features
>>
>> % perform train and test split
>> cv = cvpartition(height(T), 'HoldOut', 0.3);
>> idxTrain = training(cv);
>> idxTest = test(cv);
>> XTrain = X(idxTrain,:);
>> YTrain = Y(idxTrain);
>> XTest = X(idxTest,:);
>> YTest = Y(idxTest);
```

Training a Decision Tree (DT) Classifier

```
>> % Train Decision Tree Classifier
>> treeModel = fitctree(XTrain, YTrain, "MaxNumSplits",30); % MaxNumSplits set to 30 to increase model sensitivity and control overfitting
>>
>> % visualise Tree model
>> view(treeModel, 'Mode', 'graph');
>> Y = categorical(T.ASTHMA_WORSENER); % convert to categorical feature
>> view(treeModel, 'Mode', 'graph');
--
```

Evaluate DT Model

```
>> % evaluate model
>> YPred=predict(treeModel, XTest);
>> confusionchart(YTest,YPred);
>> title('Decision Tree - Asthma classification');
>>
>> % Check accuracy
>> accuracy = mean(YPred == YTest);
>> disp(['Accuracy: ', num2str(accuracy)]);
Accuracy: 0.86233
>>
>> % Create an ROC graph (AUC) for Decision tree classifier
>> [y_pred_tree, scores_tree] = predict(treeModel, XTest); % predicted class scores-not predicted classes
>> posScores_tree = scores_tree(:,2); % score_tree contains probability of class 0 or 1; prob of class 1 preferred
>>
>> % computing ROC points and AUC; Using perfcurve
>> [Xroc_tree, Yroc_tree, T_tree, AUC_tree] = perfcurve(YTest, posScores_tree, 1);
>>
>> % plot the ROC curve
>> figure
>> plot(Xroc_tree, Yroc_tree, 'LineWidth', 2)
>> hold on
>> plot([0 1], [0 1], '--') % random classifier baseline
>> xlabel('False Positive Rate')
>> ylabel('True Positive Rate')
>> title(['Decision Tree ROC Curve (AUC = ', num2str(AUC_tree, 3), ')'])
>> grid on
```

Train a Logistic Regression Model

```
>> % Train Logistic Regression Model
>> logModel = fitglm(XTrain, YTrainNum, 'Distribution','binomial');
>> probLog = predict(logModel, XTest);
>> YPredLog = probLog > 0.5;
```

Evaluate Logistic Regression Model

```
>> % draw log regression confusion matrix
>> YPredLogNum = double(YPredLog);
>> confusionchart(YTestNum, YPredLogNum);
>> title('Logistic Regression Matrix – Asthma Classification');
>> % check its accuracy
>> accLog = mean(YPredLog == YTestNum);
>> disp(['Logistic Regression Accuracy: ', num2str(accLog)]);
Logistic Regression Accuracy: 0.86233

>> % Compute ROC/AUC graph for Logistic Regression model
>> posScores_logreg = predict(logModel, XTest); % Get predicted probabilities – not predicted classes/labels
>>
>> % Compute ROC points and AUC; Using perfcurve
>> [Xroc_log, Yroc_log, T_log, AUC_log] = perfcurve(YTest, posScores_logreg, 1);

>> % Plot ROC curve
>> figure
>> plot(Xroc_log, Yroc_log, 'LineWidth', 2)
>> hold on
>> plot([0 1], [0 1], '--') % random baseline
>> xlabel('False Positive Rate')
>> ylabel('True Positive Rate')
>> title(['Logistic Regression ROC Curve (AUC = ', num2str(AUC_log, 3), ')'])
>> grid on
```

Train a Random Forest Model

```
>> % Train Random Forest Model
>> % Change target to categorical
>> Y = categorical([T.ASTHMA_WORSENEDE]);
>> % Derive your train and test split
>> cv = cvpartition(Y, 'HoldOut', 0.3);
>> XTrain = X(training(cv),:);
>> YTrain = Y(training(cv));
>> XTest = X(test(cv),:);
>> YTest = Y(test(cv));
```

```

>> % Address class imbalance
>> countcats(YTrain)

ans =

    6043
     957

>> % Compute class weights
>> classCounts = countcats(YTrain);
>> classWeights = sum(classCounts) ./ classCounts;

>> % assign weights per observation
>> W = zeros(size(YTrain));
>> W(YTrain == '0') = classWeights(1);
>> W(YTrain == '1') = classWeights(2);

>> % Train Random Forest Model
>> numTrees = 300;
>> rfModel=TreeBagger(numTrees, XTrain, YTrain, Method="classification",
    OOBPrediction="on", MinLeafSize=10, NumPredictorsToSample="all",Weights=W);

```

Evaluate RF Model

```

>> % Evaluate model
>> [YPred, scores] = predict(rfModel, XTest);
>> pYes = scores(:, strcmp(rfModel.ClassNames, 'Yes')); % exact probability of worsening
>>

>> % Confusion matrix
>> YPred = categorical(YPred);
>> class(YTest)

ans =

    'categorical'

>> class(YPred)

ans =

    'categorical'

>> confusionchart(YTest, YPred)
>> title('Random Forest - Confusion Matrix');

```

```

>> % Calculate accuracy
>> accuracy = mean(YPred == YTest);
>> disp(['Accuracy: ', num2str(accuracy)]);
Accuracy: 0.791

>> % Plot ROC (AUC) curve
>> rfModel.ClassNames

ans =

2x1 cell array

    {'0'}
    {'1'}

>> pYes = scores(:, strcmp(rfModel.ClassNames, '1'));
>> class(pYes)

ans =

'double'

>> [Xroc, Yroc, T, AUC] = perfcurve(YTest, pYes, '1');
>> figure
>> plot(Xroc, Yroc, 'LineWidth', 2)
>> plot([0 1], [0 1], '--')
>> plot(Xroc, Yroc, 'LineWidth', 2)
>> xlabel('False Positive Rate')
>> ylabel('True Positive Rate')
>> title(['Random Forest ROC (AUC = ' num2str(AUC) ')'])
>> grid on
>>

```